Trust, Distrust and Committed Dynamics for Sybil-Resilient Communities in Social Graphs

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Abstract

Sybil-resilient community formation builds communities of genuine identities on graphs of trust or mutual sureties. Viewed as a digital social contract, it lets identities say only one thing to each other: "I trust you". We study the next two layers of such a contract. The second layer adds distrust edges, by which an identity refuses to share a community with another. We show where the existing machinery of trust-graph connectivity, conductance and vertex expansion carries over to signed graphs, where it breaks, and how it can be strengthened: a signed conductance that subtracts distrust across a cut preserves every safety guarantee, and, when honest identities register distrust toward byzantine ones, tightens them, up to excluding byzantine members altogether. The third layer adds commitments that identities make in advance about how to respond when others disagree: taking sides, severing trust, abstaining, propagating distrust. These commitments turn a single declaration into a dynamic process on the social graph. If identities caught between the two parties must stop trusting one of them, loyal honest identities quarantine any byzantine party at no cost in honest trust. Declaring distrust toward the dropped party, rather than merely dropping it, turns a one-round local adjustment into a cascade along triangles that drives the graph toward structural balance, and that, under loyalty, only ever targets byzantine identities. Bounding the radius of obligations confines every dispute to a ball around its parties, at the price of a local quarantine; allowing abstention creates firebreaks that stop disputes and cascades, costs the abstainer, and gives the adversary nothing. Technical details and proofs are in the appendix.

1 Introduction

A digital social contract [3] is a voluntary agreement among genuine persons, written as a program: it specifies which signed acts the parties may perform and what they commit to do. The trust-graph approach to sybil-resilient community formation [1] and its mutual-sureties counterpart [2] can be read as a basic contract of this kind. Its social acts are positive: identities declare trust or pledge sureties, and a community admits a candidate only while the trust graph induced on the community remains well connected, as measured by conductance or vertex expansion. Under assumptions on how often honest identities trust byzantine ones, this keeps the fraction of byzantine members below a target.

The basic contract has no way to say no. The absence of trust means either "I do not know you" or "I object to sharing a community with you", and the contract cannot tell them apart. We argue that the natural next act is the distrust edge, and that it serves sybil resilience in three ways. First, it makes communities consensual: members can object to a candidate, not only vouch for one. Second, it records information: people notice identities that behave like sybils or their operators, and distrust is where that observation goes. Third, it costs the protocol nothing in truth conditions:

unlike an accusation, a distrust edge is a preference about co-membership, not a claim about the other identity, so it cannot be false.

Once distrust exists, a contract can also specify what others do about it. When u distrusts v, identities who trust both are caught in the middle. The contract may commit them, in advance, to take a side, to sever trust with the other side, to abstain, or to propagate distrust. Such commitments are bilateral promises, but their consequences are collective: a single declaration sets off a process on the whole social graph.

We present the contract in three layers.

1. **Trust** (Section 2). The contract of [1, 2], which we use as is.
2. **Distrust** (Section 3). We audit the machinery of the first layer on signed graphs. Most of it carries over when computed on trust edges alone; reading distrust as connectivity breaks it; and a signed conductance that subtracts distrust across cuts is both safe and, under an awareness assumption, stronger than its unsigned counterpart. We identify the genuinely new problems distrust raises: liveness, exit, and the loss of a single growing community.
3. **Commitments** (Section 4). We explore the dynamics induced by commitments to take sides, and by three modifiers: propagating distrust, bounding the radius of obligations, and allowing abstention. We contrast the local commitment with its propagating version, show how the conflict commitment quarantines byzantine parties, and show what radius and abstention control.

Proofs and formal definitions are deferred to the appendix.

2 Layer 1: the trust contract

We build on [1] and [2] and do not restate their results. We use their vocabulary. Identities are *genuine* or *sybil*; a genuine identity is *corrupt* if it ever trusts a sybil, and *honest* otherwise; corrupt identities and sybils together are *byzantine*. *Attack edges* are trust edges between honest and corrupt identities, and no trust edge joins an honest identity to a sybil. A community's *byzantine fraction* is the share of its members that are byzantine.

The guarantees of [1] take the following shape. A community grows by admitting candidates one at a time. If the trust graph induced on the community has bounded degrees and high conductance (no set of members has few trust edges to the rest relative to its volume), and attack edges are scarce (honest members have few trust edges to byzantine members), then the byzantine fraction stays below a target along the growth; a variant uses vertex expansion and a bound on the fraction of corrupt identities. In [2], identities pledge mutual sureties of increasing strength on a ledger of genuine personal identifiers; an analogous conductance theorem holds for surety graphs, and a stochastic model in which detected sybils are expelled with their connected component bounds sybil penetration.

In digital-social-contract terms the basic contract has three acts: mutual Trust, which requires both signatures; unilateral Untrust; and Admit, whose precondition is the connectivity condition (Appendix A).

3 Layer 2: distrust

The second layer adds one unilateral act: $\mathsf{Distrust}(x \to y)$, by which x declares that it refuses to share a community with y. It withdraws any trust between the two and blocks renewed trust until

revoked. The social graph becomes a signed graph: trust edges are positive, distrust edges negative. A distrust edge is authored by one identity but binds both, since co-membership is symmetric. We write Γ^+ for the *positive part*, the graph of trust edges alone.

3.1 What carries over

Every safety argument of [1, 2] is a statement about the positive part and the identity types. Two choices make them carry over unchanged.

Types ignore distrust. An identity is corrupt if it ever trusts a sybil. If distrust counted, an honest identity that distrusts a sybil would become corrupt, punishing exactly the behaviour the contract should encourage. In [2], types are defined by ledger declarations and are unaffected.

Connectivity on the positive part. If degrees, volumes, conductance, vertex expansion, spectral bounds and connected components are computed on Γ^+, the positive part with the type partition is a trust graph (or surety graph) in the original sense. Distrust then acts only through which admissions are allowed, and the theorems of [1, 2] hold for every pattern of distrust (Proposition 22).

3.2 What breaks

Distrust read as connectivity. If distrust edges were counted as edges, three things go wrong (Appendix B.1). The vertex-expansion guarantee of [1] fails outright: its proof relies on the absence of edges between honest identities and sybils, and an honest identity distrusting a sybil creates one. A four-identity example satisfies every hypothesis and still exceeds the target byzantine fraction (Example 23). Honest distrust toward byzantine members would count against attack-edge scarcity, so noticing sybils would look like trusting them. And byzantine identities could raise measured conductance at will by declaring distrust toward honest ones. In the stochastic model of [2], the connected component of a detected sybil would contain the honest identities that distrust it. The rule is therefore: *distrust never counts as connectivity.*

Monotone histories. The guarantees of [1] are stated for communities that only grow. Distrust makes membership shrink: declaring distrust withdraws trust and lowers degrees, and a community may have to expel or lose members to respect distrust. Safety survives as long as each step changes the community by a bounded amount, by a simple counting argument that separates the single-community bound from the history (Lemma 24).

Liveness. The original analyses assume that growth eventually happens. Distrust gives the adversary a way to stop it, and how much power it gets depends on how distrust is counted at admission. We consider a *hard* rule (no distrust between candidate and members) and a *soft* rule (distrust at most a fraction τ of the candidate's trust edges into the community), each either *global* (all distrust counts) or *scoped* (only distrust between identities sharing a trust neighbour counts).

Proposition 1 (Liveness under admission rules)**.** *Under the global hard rule, one byzantine member can block every candidate. Under the global soft rule, the byzantine members can block every candidate once they outnumber τ times the maximum degree, which happens once the community is large enough. Under either scoped rule, byzantine distrust can affect only honest candidates within distance two of an attack edge, at most $g(d+1)$ identities for g attack edges and maximum degree d.*

Exit. When distrust arises between members, the contract must say who leaves. The natural options behave very differently (Proposition 25): if the distrusted member leaves, one byzantine member can expel every honest member; if the author leaves, honest members who notice byzantine ones are the ones pushed out; if a member leaves only when a quorum of the members sharing a trust neighbour with it distrust it, the adversary can expel only honest members within distance two of an attack edge.

One community. Both [1] and [2] grow a single community toward the whole honest population. With honest disagreement, no single acceptable community may contain all honest identities. The natural object becomes a partition into acceptable communities, each sybil-resilient, and neither original analysis addresses partitions.

3.3 Using distrust: signed conductance

Computing everything on the positive part makes distrust harmless, but also useless to the connectivity machinery. A signed measure can do better, as long as distrust can only lower it. Define the *net conductance* of a community as its conductance with the distrust edges across each cut subtracted from the trust edges across it:

$$\Phi^{\pm}(A) \;=\; \min_{S} \; \frac{e^{+}(S, A \setminus S) - e^{-}(S, A \setminus S)}{\min\big(\mathrm{vol}^{+}(S), \mathrm{vol}^{+}(A \setminus S)\big)},$$

with volumes taken in the positive part. Distrust across a cut signals that the two sides do not belong together, which is the opposite of what trust across it signals.

Theorem 2 (Signed conductance)**.**

1. Safety. *Net conductance never exceeds positive conductance, so replacing conductance by net conductance in the admission condition of [1] preserves every safety guarantee. More generally, any signed connectivity measure bounded above by its positive counterpart is safe; any measure that can exceed it, such as the unsigned one, is not.*

2. Strengthening. *Suppose honest members register, for each attack edge inside the community, on average at least $\kappa < 1$ distrust edges toward byzantine members (awareness). Then the conductance threshold of [1] that guarantees a byzantine fraction below β can be lowered by the factor $1 - \kappa$ when net conductance is used, provided the byzantine members do not hold more trust volume than the honest ones.*

3. Exclusion. *If awareness holds with $\kappa \geq 1$, any community whose members all have trust edges inside it and whose net conductance is positive is either entirely honest or entirely byzantine.*

The intuition for (2) and (3) is the cut separating the byzantine members from the honest ones. Its trust edges are exactly the attack edges inside the community; awareness puts at least κ distrust edges across it for each of them. Net conductance at that cut is therefore at most $1 - \kappa$ times the positive one, and becomes non-positive when $\kappa \geq 1$.

Net conductance has a price. Honest disagreement across a cut lowers it as well, so a community with internal friction must be better connected to grow, or split. Hostile distrust declared by byzantine members lowers it too: it can stall growth but never weakens the guarantee. A second way to use distrust replaces the scarcity hypothesis itself: under awareness, the observable density of distrust inside a community bounds the unobservable number of attack edges, giving a fully checkable admission condition (Lemma 26).

Table 1 summarizes the audit.

Building block of [1, 2]	With distrust edges
Identity types	Carry over if defined by trust only
Conductance, vertex expansion, spectral bounds	Carry over on the positive part; fail if distrust counts as an edge
Signed (net) conductance	Safe; strengthened by awareness; excludes byzantine members when awareness is strong
Attack-edge scarcity	Can be certified by internal distrust density, under awareness
Sybil-component expulsion [2]	Components must be computed on the positive part
Growing (nested) histories	Replaced by bounded churn
Eventual growth (liveness)	Lost under global admission rules; restored near attack edges only by scoping
Single community	Replaced by partitions (open)

Table 1: The machinery of trust-based community formation on signed graphs.

4 Layer 3: commitments and their dynamics

4.1 Disputes and the actions of agents

A declaration of distrust between u and v opens a *dispute* with *parties* u and v. Each party starts a *side*. An identity outside both sides that trusts a member of a side is *exposed* to it; one exposed to both sides is *conflicted*: it trusts two identities that are now on opposite sides. An identity caught in a dispute has four possible actions.

- *Side*: join one side and withdraw trust from the members of the other side it trusts.
- *Abstain*: withdraw trust from the members of both sides it trusts, and join neither.
- *Wait*: do nothing for now.
- *Declare*: when withdrawing trust while taking a side, also declare distrust toward the identities dropped.

A *commitment* is a clause that identities accept in advance and that fixes four things: *who must act* (and which actions are allowed to them), *how far* obligations reach, whether *abstention* is allowed, and whether withdrawing trust comes with *declaring* distrust. We study two basic commitments and three modifiers.

- *Local commitment.* Only the common trust neighbours of u and v act: each drops one of its two trust edges to the parties. Sides do not grow.
- *Conflict commitment.* Every conflicted identity must take a side. An identity exposed to one side only may join it, or wait. Sides grow, and whoever joins a side can create new conflicts for its own trust neighbours.
- *Propagation* (modifier). Withdrawing trust while taking a side comes with declaring distrust toward each identity dropped. Each declaration is a new dispute.
- *Bounded radius* (modifier). Only identities within trust distance r of the parties take part: nobody farther away is obliged, or allowed, to act.

- *Abstention* (modifier). A conflicted identity may abstain instead of taking a side. Abstention withdraws trust but declares nothing.

Each commitment is a promise an identity makes to its own trust neighbours. The local commitment says "if you and someone else I trust fall out, I will drop one of you". The conflict commitment says "if two people I trust end up on opposite sides, I will not keep trusting both".

All variants share a basic structure (Lemma 27): trust is withdrawn only from a side member, never between two identities outside the sides; no trust edge remains between the two sides; each side stays connected; and every dispute settles after finitely many actions.

4.2 Local commitment versus propagation

The local commitment and its combination with propagation look alike at the level of a single agent: a common neighbour w of u and v drops one of them. They differ in one act, and that act changes the dynamics entirely.

Actions. Under the local commitment, w drops its trust in, say, v. The edge simply disappears: w now neither trusts nor distrusts v. Under propagation, w also declares distrust toward v. The trust edge is not removed but reversed: w now actively refuses to share a community with v.

Dynamics. Under the local commitment, a dispute ends in one round. Only the common neighbours of the two parties act, nobody else is affected, and no new dispute arises. Under propagation, every dropped edge becomes a new dispute between w and the party it dropped, whose common neighbours must act in turn. The result is a cascade that travels along triangles of the trust graph and can reach far from the original parties.

Theorem 3 (Local cascade)**.** *Under the local commitment with propagation, the cascade started by any set of declarations ends after at most one dispute per declaration and per original trust edge. At the end, every original trust edge is either still trust or has become distrust, and no triangle has exactly one distrust edge. If the graph of trust and declared distrust is chordal, the result is clusterable in the sense of Davis [8]: the trust graph splits into components with all distrust between them, and in particular every pair in distrust is separated.*

Consequences. Table 2 summarizes the contrast. The local commitment *neutralizes*: it removes a few trust edges, records nothing, and leaves the parties connected; on a long cycle nothing happens at all. Propagation *polarizes*: every withdrawn trust edge becomes distrust, the cascade drives the signed graph toward structural balance (exactly on chordal graphs, triangle by triangle in general), and what began as one disagreement ends as a pattern of opposing camps. In the sybil setting, propagation is also what turns side-taking into information.

Corollary 4 (Truthful cascades)**.** *Suppose a cascade starts from a dispute between an honest and a byzantine party, and honest identities always keep the honest party when they must choose between an honest and a byzantine one (loyalty, Assumption 1). Then the cascade never withdraws trust between two honest identities, and every distrust edge declared by an honest identity targets a byzantine one.*

The reason is that a new dispute is opened by the identity that dropped a party, against that party. A loyal honest identity drops only byzantine parties, so the cascade never pits two honest identities against each other.

	Local commitment	Local commitment with propagation
Action of a common neighbour	Drops trust in one party	Drops trust in one party and declares distrust toward it
Edge afterwards	Absent (neutral)	Negative (opposition)
New disputes	None	One per dropped edge
Who acts	Common neighbours of the parties	Common neighbours of every pair in a dispute, along triangles
Duration	One round	Up to one dispute per trust edge
End state	Parties share no trust neighbour	No triangle with exactly one distrust edge; balanced on chordal graphs
Separation of the parties	No	On chordal graphs
Information recorded	None	Distrust edges; truthful under loyalty

Table 2: The local commitment with and without propagation.

4.3 The conflict commitment: quarantine

Under the conflict commitment, an identity that takes a side becomes a member of it, so its own trust neighbours may in turn become conflicted. Sides grow through the trust graph, but only as far as conflicts and voluntary side-taking carry them. Consider a dispute between an honest party h and a byzantine party b, whichever of them declared the distrust.

Assumption 1 (Loyalty)**.** In a dispute between an honest and a byzantine party, honest identities never join the byzantine party's side, and join the honest party's side whenever they are exposed to it.

Theorem 5 (Quarantine)**.** *Under the conflict commitment and loyalty, when the dispute settles:*

1. *no honest identity is on the byzantine party's side;*
2. *no trust between two honest identities has been withdrawn;*
3. *every honest identity connected to h through honest identities is on h's side;*
4. *the byzantine party, and every identity that sided with it, keeps no trust edge to any of these honest identities.*

The adversary faces a dilemma at every byzantine identity that trusts both sides: abandon the byzantine party, or abandon its attack edges. It can keep a path between the parties only through byzantine identities that never take a side. The theorem does not depend on who started the dispute, so a sybil that declares distrust toward an honest identity quarantines itself.

Corollary 6 (Truthful propagation)**.** *Under the conflict commitment, loyalty and propagation, every distrust edge that an honest identity declares during the dispute targets a byzantine identity.*

This closes a loop with Layer 2: the commitments generate exactly the honest distrust toward byzantine identities that the awareness assumption of Theorem 2 and Lemma 26 postulates. Loyalty is itself a behavioural assumption of the same kind as awareness.

Unlike the quarantine guarantee, the reach of an honest dispute under the conflict commitment is not bounded a priori: a chain of identities who each take a side can carry conflicts far from the parties. The next two modifiers control this reach in different ways.

4.4 Bounded radius: reach

With bounded radius r, only identities within trust distance r of the parties take part. Inside this ball the commitment works as before; outside it nothing happens.

Proposition 7 (Confinement)**.** *Under any commitment with radius r, every action and every withdrawn trust edge lies within distance r of the parties. In a trust graph of maximum degree d, at most $2\big(1+\sum_{i=1}^{r} d(d-1)^{i-1}\big)$ identities take part. For the local cascade, every distrust edge it creates lies in the ball, and no triangle with its third vertex in the ball has exactly one distrust edge.*

The price is reach. The parties are no longer separated even under conditions that would separate them otherwise: they typically stay connected through paths that leave the ball. The quarantine becomes local.

Theorem 8 (Bounded quarantine)**.** *Under the conflict commitment with radius r and loyalty, statements (1) and (2) of Theorem 5 hold, and (3) and (4) hold for the honest identities connected to h through honest identities within the ball. The byzantine party may keep attack edges to honest identities outside the ball, or inside it but reachable from h only through paths that leave it.*

Bounded radius therefore trades the reach of a quarantine against the damage of honest disputes, which under the conflict commitment are otherwise limited only by how far side-taking spreads. For quarantine the trade is mild: under loyalty no honest trust is lost at any radius, so radius only limits how many attack edges the byzantine party loses. For honest disputes it is the only a-priori bound on how much trust is withdrawn.

4.5 Abstention: firebreaks

An abstaining identity withdraws its trust from the members of both sides it trusts, joins neither, and declares nothing. Because it is on no side, it exposes nobody: abstention stops conflicts, and cascades, from passing through it. If it later becomes conflicted again because neighbours have taken opposite sides, it must again choose or abstain.

Proposition 9 (Containment)**.** *With abstention allowed, every dispute settles after at most one side-taking per identity and at most one abstention per two trust edges. If every conflicted identity abstains and nobody takes a side voluntarily, the dispute settles in one round: exactly the common neighbours of the parties act, each withdrawing both of its trust edges to them. In the local cascade, an abstaining identity opens no new dispute.*

Abstention is neutrality at a price: the abstainer loses trust on both sides. It does not help the adversary.

Proposition 10 (Abstention and the adversary)**.** *Under the conflict commitment with abstention, suppose honest identities never join the byzantine party's side but may abstain. Then no honest identity is on the byzantine side, and every trust edge withdrawn has a byzantine endpoint or an honest abstainer as endpoint. If honest identities are fully loyal, they never abstain and Theorem 5 holds unchanged. A byzantine identity that abstains loses its trust edges to both sides, attack edges included.*

For the adversary, abstention is the worst of the three options. For honest identities it is the safe option when they cannot tell which party is honest: it costs them trust but never places them on a byzantine side. For honest disputes it is the strongest damage control available, because it bounds the spread of a dispute without requiring anyone to judge between the parties.

Behaviour	Who acts	Dynamics	Guarantee
Local commitment	Common neighbours of the parties	One round; trust edges vanish	No separation; minimal damage
+ propagation	Common neighbours of every disputed pair	Cascade along triangles; trust turns into distrust	Triangle balance; full balance on chordal graphs; truthful under loyalty
Conflict commitment	Conflicted identities; others may join	Sides grow as far as conflicts carry them	Quarantine under loyalty
+ propagation	Same	Same; withdrawn trust recorded as distrust	Honest distrust targets byzantine identities under loyalty
+ radius r	Only within distance r	Confined to the ball	Local quarantine; bounded damage
+ abstention	Conflicted identities may abstain	Abstainers stop conflicts and cascades	Adversary gains nothing; honest abstainers pay

Table 3: Dynamics induced by the commitments and modifiers.

4.6 Summary of the dynamics

Table 3 summarizes the dynamics induced by each behaviour.

5 Sybil Resilience through Trust and Distrust

We specify a mechanism for separating persons who distrust each other into different components of the friendship graph. Friendship represents the mutual sureties of the social graph [2]. The mechanism resolves declared distrust; determining which identifiers are sybils remains a separate task. Following [4], we first state two grassroots contracts and then their speech-act schemas, which denote volition-guarded atomic transactions [5].

5.1 Grassroots Contracts

Each person freely signs the following undertaking towards all other signatories and presents it to every person with whom they act under it. All signatories have the same rights and obligations. No external authority grants permission to enter a relationship, and no common register of people is required. The exchange is of mutual sureties and reciprocal commitments to respect these rules.

GD: Grassroots Distrust

GD extends GSG [4].
Alice, Bob and Carol are persons.

1. ■ *Distrust.* Alice, who knows Carol and does not distrust Carol, may distrust Carol; if they are friends, this ends their friendship.
2. ■ *Propagated distrust.* Alice, who is not a friend of Carol and does not distrust Carol, and whose friend Bob distrusts Carol, may distrust Carol.
3. ■ *Ceasing to distrust.* Alice, who distrusts Carol, may cease to distrust Carol.

Pragmas

Distrusted: the persons Alice distrusts.
Distrust: `Alice [Distrust]`
Propagated distrust: `Alice [Propagate] <person>`
Ceasing to distrust: `Alice [Cease]`

GD-S: Separating GD

GD-S restricts GD.
Alice, Bob and Carol are persons.

1. ◆ *Friendship.* GSG:1, only if Alice and Bob do not distrust each other.
2. ○ *Taking sides.* Alice, whose friend Bob distrusts Carol, will eventually distrust Carol or end their friendship with Bob.

GD-S keeps every act of GD but befriending, which GD-S:1 restricts, and adds an undertaken clause, whose formula denotes no transaction [4].

5.2 Speech-Act Schemas

The clauses of GD and GD-S:1 describe acts, and GD-S:2 is an undertaking; their speech-act schemas [4] are below. GD has the schemas of GKK and of GSG [4] and four more over one predicate: *distrusts*(*Carol*) in the state of Alice says that Alice distrusts Carol. GD-S has the schemas of GD, with *befriend* restricted. The boxes below have them by the numbers of the clauses of the two contracts. A schema is written with its head before the arrow and its body after. The first argument of an atom is its holder; the body is what the act requires of the machines and, negated, forbids; the head is what it adds and, negated, deletes; and the guard names the persons who must will it. Every role requires *identity*, which is not shown.

GD: Speech-Act Schemas

1. $\mathit{distrusts}(\mathit{Alice}, \mathit{Carol}) \leftarrow$

   $\mathit{knows}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{friend}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{distrusts}(\mathit{Alice}, \mathit{Carol})$
   guarded by *Alice*.

   $\mathit{distrusts}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{friend}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{friend}(\mathit{Carol}, \mathit{Alice}) \leftarrow$

   $\mathit{knows}(\mathit{Alice}, \mathit{Carol})$, $\mathit{friend}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{distrusts}(\mathit{Alice}, \mathit{Carol})$, $\mathit{friend}(\mathit{Carol}, \mathit{Alice})$ guarded by *Alice*.

2. $\mathit{distrusts}(\mathit{Alice}, \mathit{Carol}) \leftarrow$

   $\mathit{friend}(\mathit{Alice}, \mathit{Bob})$, $\neg\mathit{friend}(\mathit{Alice}, \mathit{Carol})$, $\neg\mathit{distrusts}(\mathit{Alice}, \mathit{Carol})$, $\mathit{distrusts}(\mathit{Bob}, \mathit{Carol})$ guarded by *Alice*.

3. $\neg\mathit{distrusts}(\mathit{Alice}, \mathit{Carol}) \leftarrow$

   $\mathit{distrusts}(\mathit{Alice}, \mathit{Carol})$ guarded by *Alice*.

GD-S: Speech-Act Schemas

1. $\mathit{friend}(\mathit{Alice}, \mathit{Bob})$, $\mathit{friend}(\mathit{Bob}, \mathit{Alice}) \leftarrow$

   $\mathit{knows}(\mathit{Alice}, \mathit{Bob})$, $\neg\mathit{friend}(\mathit{Alice}, \mathit{Bob})$, $\neg\mathit{distrusts}(\mathit{Alice}, \mathit{Bob})$, $\mathit{knows}(\mathit{Bob}, \mathit{Alice})$, $\neg\mathit{friend}(\mathit{Bob}, \mathit{Alice})$, $\neg\mathit{distrusts}(\mathit{Bob}, \mathit{Alice})$ guarded by Alice, Bob.

2. $\mathit{friend}(\mathit{Alice}, \mathit{Bob}) \wedge \mathit{distrusts}(\mathit{Bob}, \mathit{Carol}) \wedge \neg\mathit{distrusts}(\mathit{Alice}, \mathit{Carol}) \rightsquigarrow$

   $\mathit{distrusts}(\mathit{Alice}, \mathit{Carol}) \vee \neg\mathit{friend}(\mathit{Alice}, \mathit{Bob})$ undertaken by Alice.

GD:1 has two schemas: *distrust*, of a person Alice knows who is not their friend, and *distrust_friend*, of a friend, which ends the friendship. GD:2 is *propagate*: Bob neither wills the act nor changes state, and the act reads Bob's record of distrusting Carol. GD:3 is *cease*. A person distrusts another by an act of their own: no act adds a distrust to the state of a person who does not will it. GD-S:1 is *befriend*, which forbids at each of the two a distrust of the other; it has the name and the guards of GSG's *befriend* and a stronger body. GD and GD-S have no schema with no guarding agent role, so they have simple unguarded classes, and GD-S restricts GD [4, Proposition Syntactic restrictions]: every correct run of *GD-S* is, on the machines, one of *GD*. An undertaken clause leaves the schemas and the runs as they are [4]. GD-S:2 is undertaken. It has an undertaking formula [4], which denotes no transaction and is a condition on runs, that whenever its condition holds its response eventually holds. Alice keeps it by distrusting Carol or by ending their friendship with Bob, acts of GSG and GD which Alice alone wills.

Each schema denotes volition-guarded transactions among agents, an agent being a person with a will and a machine with a state, and a contract denotes the protocol over them [4, 5]; we write *GD* and *GD-S* for the two protocols. A run of a protocol is *correct* if it is safe and live: it takes only transactions of the protocol, and every class of transactions that stays enabled is eventually taken [5]. A person changes what they will at any time, with no liveness obligation.

Proposition 11 (Grassroots property)**.** *GD and GD-S are syntactically grassroots [4]. Hence the protocols GD and GD-S are grassroots and volitionally grassroots [4, Theorem Grassroots realisation].*

Proof. GSG is syntactically grassroots [4]. GD extends GSG, so by [4, Theorem What the check preserves] it suffices that GD has an unobstructed introductory act, that every atom an added schema adds has traceable provenance in GD, that every added schema satisfies volition in GD, and that every kept schema with no guarding role and a reading role has a signature atom; GKK and GSG have no schema with no guarding role. *Introduction: acquaint* is the unobstructed introductory act of GKK [4]; it forbids nothing, and no schema of GD adds or deletes *identity*, the atom it requires, but *establish*, so it remains one in GD. *Provenance:* the atom $\mathit{distrusts}(\mathit{Carol})$ is added at Alice: by *distrust*, where Alice requires $\mathit{knows}(\mathit{Carol})$; by *distrust_friend*, where Carol is a role; and by *propagate*, where the role Bob requires $\mathit{distrusts}(\mathit{Carol})$. *Volition: distrust* and *cease* have one role; in *distrust_friend* the guarding role Alice requires $\mathit{friend}(\mathit{Carol})$, an atom of traceable provenance naming the other role; and in *propagate* the guarding role Alice requires $\mathit{friend}(\mathit{Bob})$, naming the reading role Bob. GD-S is obtained from GD by replacing *befriend* by a schema with its name, head and guards and a stronger body, with no new role, so by the same theorem it suffices that GD-S has an unobstructed introductory act, which *acquaint* remains: GD-S keeps it and adds no schema that adds or deletes *identity*. The three conditions do not read the formula of GD-S:2 [4]. □

5.3 The Friendship and Distrust Graph

We use $I \subset \Pi$ for a finite nonempty set of agents; different agents need not have different persons.

Definition 12 (Friendship and distrust graph)**.** Let m be a machine configuration over I. For $x \in I$, F_x is the set of agents y with *friend*(y) in the state of x, and B_x the set of agents q with *distrusts*(q) in the state of x. The ***friendship edges*** and the ***distrust arcs*** of m are

$$T(m) = \{\{x, y\} : x, y \in I,\ y \in F_x\},$$
$$D(m) = \{(x, q) : x, q \in I,\ q \in B_x\},$$

and the ***graph*** of m is the pair $(T(m), D(m))$. The configuration is ***well formed*** if, for every $x, y \in I$, $y \in F_x$ implies $x \in F_y$, $y \notin B_x$ and $x \notin B_y$. It is ***separated*** if for every distrust arc (x, q), x and q are in different connected components of $(I, T(m))$.

The graph is a view of the agents' local records; the protocol has no global store. A friendship is recorded by both friends, and a distrust by the agent who distrusts. In GD a person may befriend a person they distrust, so a configuration of GD need not be well formed; GD-S:1 forbids it.

Lemma 13 (Well-formedness)**.** *In every safe run of GD-S, every machine configuration is well formed.*

Proof. By induction on the run. Initially all states are empty. *befriend* adds the friendship at both agents and requires that neither distrusts the other. *unfriend* and *distrust_friend* delete the friendship at both. *distrust* and *propagate* add a distrust of Carol at Alice and require that Carol is not a friend of Alice, so by the induction hypothesis Alice is not a friend of Carol. *cease* deletes a distrust. The other transactions and the volition changes change no *friend* and no *distrusts* atom. □

5.4 Taking Sides

Definition 14 (Obligation)**.** For distinct $x, p, q \in I$ and a machine configuration m,

$$\mathsf{obl}_m(x, p, q) \iff p \in F_x, \quad q \in B_p, \quad q \notin B_x.$$

A configuration is ***quiescent*** if no obligation holds in it.

The obligation $\mathsf{obl}_m(x, p, q)$ is the condition of the undertaking formula of GD-S:2, with x, p and q for Alice, Bob and Carol.

Definition 15 (Keeping and breaching GD-S:2)**.** Agent x ***breaches*** GD-S:2 in a run if, for some p and q, the obligation $\mathsf{obl}_m(x, p, q)$ holds in a configuration of the run, and in every later configuration $p \in F_x$ and $q \notin B_x$; x ***keeps*** GD-S:2 in the run otherwise [4]. A finite run is regarded as remaining in its last configuration.

An agent can always keep GD-S:2. While $\mathsf{obl}_m(x, p, q)$ holds, the person of x may will a *response*: either *unfriend* of x and p, or a distrust of q, which is *distrust_friend* of x and q if $q \in F_x$, and *propagate* of x, p and q otherwise. In a well-formed configuration the machine preconditions of each hold, and each is guarded by x, so in a correct run a response that x wills, and goes on willing, is taken, and the response of the formula then holds. Neither p nor q can prevent the response of x by withholding their will.

A breach of GD-S:2 is not a fault of the execution. A run in which a person never responds is a correct run, in which that person has failed to uphold their undertaking.

5.5 Stabilisation and Separation

Proposition 16 (Separation)**.** *Every well-formed, quiescent configuration is separated.*

Proof. Suppose $(p, q) \in D$ and that (I, T) has a simple path

$$q = v_0, v_1, \ldots, v_k = p.$$

Let $j \geq 1$ be the first index with $q \in B_{v_j}$; it exists since $q \in B_p$. The agent v_1 is a friend of q, so by well-formedness $q \notin B_{v_1}$ and $j \geq 2$. Set $x = v_{j-1}$. Then $x \neq q$, $v_j \in F_x$, $q \in B_{v_j}$ and $q \notin B_x$, so $\mathsf{obl}_m(x, v_j, q)$ holds, and the configuration is not quiescent. □

Consider a safe run of *GD-S* over n agents, and a suffix of it that begins with graph (T_*, D_*) and in which no *befriend* and no *cease* is taken.

Theorem 17 (Stabilisation)**.** *The graph changes at most $|T_*| + n^2 - |D_*|$ times in the suffix. If every agent keeps GD-S:2 in the run, the last value of the graph is quiescent and separated.*

Proof. Put $\Phi(m) = |T(m)| + n^2 - |D(m)|$. In the suffix, *unfriend* decreases Φ by one, *distrust* and *propagate* by one, and *distrust_friend* by two; every other transition leaves the graph unchanged. Since $\Phi \geq 0$, the graph changes at most $\Phi(m_*) = |T_*| + n^2 - |D_*|$ times, and is constant from some configuration on.

Suppose $\mathsf{obl}_m(x, p, q)$ holds in the constant graph. Then $p \in F_x$ and $q \notin B_x$ in every configuration from there on, and x breaches GD-S:2. Hence, if every agent keeps GD-S:2, the last value of the graph is quiescent; it is well formed by Theorem 13 and separated by Theorem 16. □

New friendships and ceasing to distrust are allowed at any time; the bound applies once they stop.

Remark 18 (The role of volition)**.** Let q and x be friends, x and p be friends, p distrust q, and no agent will any act. Agent x has the obligation $\mathsf{obl}_m(x, p, q)$, and no transaction is enabled. The configuration is reached by acquaintances and befriendings among the three and a distrust of q by p. The run that remains in it is correct, x breaches GD-S:2 in it, and it is not separated.

5.6 Resilience to Breach

Theorem 17 assumes that every agent keeps GD-S:2. For $Z \subseteq I$, a configuration is *separated on* $I \setminus Z$ if for every distrust arc (x, q) with $x, q \in I \setminus Z$, x and q are in different connected components of the friendship graph restricted to $I \setminus Z$.

Consider a safe run of *GD-S* over n agents, and a suffix of it that begins with graph (T_*, D_*) and in which no *befriend* and no *cease* is taken.

Theorem 19 (Resilience to breach)**.** *If every agent of $I \setminus Z$ keeps GD-S:2 in the run, the last value of the graph is separated on $I \setminus Z$.*

Proof. By Theorem 17 the graph is constant from some configuration on. Suppose that in the constant graph $(p, q) \in D$ with $p, q \notin Z$, and that a simple path $q = v_0, v_1, \ldots, v_k = p$ of friendship edges lies in $I \setminus Z$. The proof of Theorem 16 yields an obligation $\mathsf{obl}_m(x, v_j, q)$ with $x = v_{j-1} \notin Z$, and $v_j \in F_x$ and $q \notin B_x$ in every configuration from there on, so x breaches GD-S:2; a contradiction. □

The theorem holds for every Z: the agents who keep their undertaking are separated among themselves, however many breach it.

Remark 20 (Intermediaries in breach)**.** Separation on $I \setminus Z$ cannot be replaced by separation. In the run of Remark 18, put $Z = \{x\}$. Neither p nor q has an obligation, so both keep GD-S:2. Agent p distrusts q and the two remain in one component of the friendship graph, although the configuration is separated on $I \setminus Z = \{p, q\}$.

5.7 Sybil Repellence

Separation concerns any two agents who distrust each other. We now take the agents to be the identities of Section 2, honest or byzantine, with friendship for trust, and show that one honest agent's distrust of a byzantine agent spreads to the honest agents around them.

Assumption 2 (Loyalty in GD-S)**.** An honest agent distrusts only byzantine agents, and ends a friendship only with a byzantine agent.

An honest agent x can be loyal and keep GD-S:2. Let $\mathsf{obl}_m(x, p, q)$ hold. If p is honest then q is byzantine, since p is loyal and a distrust is added only by an act of the agent who distrusts, and x responds by distrusting q. If p is byzantine, x responds by ending their friendship with p.

Proposition 21 (Repellence)**.** *Consider a safe run of GD-S in which the honest agents are loyal and keep GD-S:2, and a suffix of it in which no befriend and no cease is taken. No friendship between two honest agents ends in the run. If an honest agent p distrusts q in the last value of the graph, then q is byzantine, and every honest agent connected to p by a path of friendships among honest agents distrusts q and is not a friend of q.*

Proof. A friendship ends by *unfriend* or by *distrust_friend*, each guarded by one of the two friends; between two honest agents neither is taken, by loyalty. A distrust is added only by an act guarded by the agent who distrusts, so by loyalty q is byzantine. By Theorem 17 the graph is constant from some configuration on. Let $p = u_0, u_1, \ldots, u_k$ be a path of friendships among honest agents in the constant graph; we show by induction on i that $q \in B_{u_i}$. This holds for $i = 0$. Let $q \in B_{u_i}$ and suppose $q \notin B_{u_{i+1}}$. The three agents are distinct, q being byzantine, so $\mathsf{obl}_m(u_{i+1}, u_i, q)$ holds, $u_i \in F_{u_{i+1}}$ and $q \notin B_{u_{i+1}}$ in every configuration from there on, and u_{i+1} breaches GD-S:2; a contradiction. By well-formedness (Theorem 13), an agent who distrusts q is not a friend of q. □

So a byzantine agent that one honest agent distrusts keeps no friendship with any honest agent of that agent's honest surroundings, at no cost in friendships among honest agents, whether or not the byzantine agents keep GD-S:2. This is the quarantine of the conflict commitment (Section 4), obtained from a contract in which each agent acts on the records of their friends. The distrust it spreads is by honest agents of byzantine ones, which is what the awareness assumption of Section 3 requires.

Neither separation nor repellence establishes that a distrust is warranted or that an identifier is a sybil; loyalty assumes that honest agents tell. Safety faults [5], in which faulty machines take transactions outside the protocol, are not addressed here.

6 Discussion

A contract for communities of genuine identities. The results point to one design: the trust contract of [1, 2], extended by distrust with (i) distrust excluded from every connectivity quantity, or used only through net conductance; (ii) scoped admission; (iii) quorum-based exit among identities sharing a trust neighbour; (iv) the conflict commitment with propagation, a radius large enough to cover the neighbourhood in which identities can judge a dispute, and abstention allowed; and (v)

distrust budgets. Its safety is that of the trust contract. The adversary's power to block or expel is confined near attack edges. A dispute between an honest and a byzantine party, whoever began it, quarantines the byzantine party from the honest identities around the dispute without loss of honest trust, and generates honest distrust aimed only at byzantine identities, which net conductance and the certified-scarcity condition convert into stronger guarantees. Honest disputes cost at most the trust inside the ball, and less where identities abstain.

The case against. The positive results rest on two behavioural assumptions, awareness and loyalty: honest identities notice byzantine ones, and side with honest parties. They are plausible locally and unverified globally. Where loyalty fails, the commitments lose their guarantee but keep one safety property: no honest identity is ever forced onto a side it does not wish to join, since abstention is always available. A second objection is that commitments bind identities to act on other people's disputes, which may sit uneasily with the voluntariness digital social contracts are meant to embody [3]. We have kept every obligation local to an identity's own trust edges; whether that suffices is a question for the formal treatment of volition in [4].

Future research. On the theory side: derive loyalty from local structure, for instance by showing that keeping the tie with more common neighbours implies loyalty when attack edges are scarce; bound how far an honest dispute spreads under the conflict commitment as a function of the trust graph; choose the radius optimally, trading the reach of quarantine against the damage of honest disputes; analyse abstention as a strategic choice, where identities weigh the trust lost by abstaining against the risk of siding with a byzantine party; characterize cascade size in terms of clustering, as a percolation problem on triangles; bound the attack edges the adversary can retain through identities that never take a side; establish liveness under scoped admission; find spectral certificates for net conductance; define sybil-resilience for partitions into acceptable communities, with connections to correlation clustering and hedonic games with friends and enemies; and express each clause as act schemas in the framework of [4]. On the empirical side, the size of cascades, the spread of honest disputes, the effect of radius and abstention, and the fragmentation caused by repeated disputes can be measured by simulation and on public signed networks [10]. Above all, awareness and loyalty can be measured in deployments of digital social contracts on grassroots platforms [6]: how often honest participants declare distrust, toward whom, which side they take when they must choose, and when they prefer to abstain.

A Contracts and notation

Signed graph. The contract state induces $\Gamma = (V, E^+, E^-)$ with undirected positive edges E^+ and directed negative edges E^-; $(x \to y) \in E^-$ means x distrusts y. We write $\Gamma^+ = (V, E^+)$, $N(x)$ for the positive neighbourhood, d for the maximum positive degree, H, C, S for honest, corrupt and sybil identities, $B = C \cup S$, g for the number of attack edges and $\partial \subseteq H$ for their honest endpoints. For a community A, $\beta(A) = |A \cap B|/|A|$; $\deg_A$, vol_A and $e^+(\cdot,\cdot)$ are computed in $\Gamma^+|_A$, and $e^-(X, Y)$ counts distrust edges between X and Y in either direction unless an orientation is written.

Acts. Following [3, 4], an act has an author, a set of parties whose signatures are required, a precondition and an effect.

- $\mathsf{Trust}(x, y)$: signatures of x and y; precondition: no distrust between them; effect: $\{x, y\} \in E^+$.

- Untrust(x, y): either party; effect: $\{x, y\} \notin E^+$.
- Distrust$(x \to y)$: author x; effect: $(x \to y) \in E^-$ and $\{x, y\} \notin E^+$. Revoke$(x \to y)$ undoes it.
- Admit(A, x): precondition: the connectivity condition of [1], and the admission rule of Section 3; Leave(A, x): author x.

Exit rules that remove a member without its consent at that time are accepted as a condition of Admit.

Admission rules. Write $e^-(x, A)$ for the distrust edges between candidate x and members of A. *Hard*: $e^-(x, A) = 0$. *Soft* with tolerance τ: $e^-(x, A) \leq \tau \deg_A(x)$. *Scoped*: only distrust edges between identities sharing a positive neighbour count.

Exit rules. *Target-exit*: the distrusted member leaves. *Author-exit*: the author leaves. *Seniority*: the later joiner leaves. *Scoped quorum* with threshold θ: a member x leaves when at least $\max(1, \theta \deg_A(x))$ members sharing a positive neighbour with x distrust it, repeated until stable.

B Layer 2: statements and proofs

Proposition 22 (Transfer)**.** *If distrust constrains only admission and all connectivity quantities are computed on Γ^+, the safety theorems of [1] and the conductance theorem of [2] hold for every set of distrust edges and every admission rule.*

Proof. Γ^+ with the type partition is a trust (or surety) graph in the original sense; every hypothesis refers only to it and to the history; an admission rule only selects which nested history is realized. □

B.1 The unsigned reading

Example 23. Let $A = \{h_1, h_2, h_3\}$ be honest with trust edges forming a triangle, and $A' = A \cup \{s\}$ with s a sybil and one distrust edge $h_1 \to s$. There are no corrupt identities. If the distrust edge is counted as an edge, the vertex expansion of A' is $\frac{1}{2}$ under either the inner or the outer boundary convention. With target $\beta = \frac{1}{5}$, corrupt-fraction bound $\gamma_v = 0.08$ and growth parameter $\delta = 0.6$, the expansion condition $\Phi_v > \gamma_v/\beta = 0.4$ of Lemma 2 of [1] holds, as do the growth and corrupt-fraction bounds, yet $\beta(A') = \frac{1}{4}$. On Γ^+, s is isolated and the admission is refused.

B.2 Churn

Lemma 24 (Counting under churn)**.** *Let A, A' be consecutive communities, $N = A' \setminus A$, $R = A \setminus A'$. If $\beta(A) \leq \frac{1-\delta}{2}$ and $|N| + |R \cap H| - |R \cap B| \leq \delta|A|$ (in particular if $|A \triangle A'| \leq \delta|A|$), then $|A' \cap B| \leq |A' \cap H|$.*

Proof. $|A' \cap B| \leq |A \cap B| - |R \cap B| + |N|$ and $|A' \cap H| \geq |A \cap H| - |R \cap H|$, so it suffices that $|N| + |R \cap H| - |R \cap B| \leq |A|(1 - 2\beta(A))$, and the right side is at least $\delta|A|$. □

Combined with the single-community form of the conductance argument of [1], which bounds $\beta(A)$ for a community with bounded degrees, scarce attack edges and high conductance provided $|A \cap B| \leq |A \cap H|$, induction gives safety for every history whose steps respect the churn bound, provided the target β is at most $\frac{1-\delta}{2}$.

B.3 Liveness and exit

Proof of Proposition 1. Hard global: the byzantine member distrusts the candidate. Soft global: each byzantine member contributes one distrust edge, and $\deg_A(x) \le d$, so $|A \cap B| > \tau d$ suffices; with byzantine fraction at least $\beta_0 > 0$ this holds once $|A| > \tau d/\beta_0$. Scoped: a byzantine distrust edge toward honest x counts only if x and its author b share a positive neighbour y. If $y \in B$ then $x \in \partial$; if $y \in H$ then $y \in \partial$ and $x \in N(\partial)$; and $|\partial \cup N(\partial)| \le g(d+1)$. □

Proposition 25 (Exit rules)**.** *(1) Under target-exit, a single byzantine member can remove every honest member. (2) Under author-exit, a rational adversary declares no distrust, and each honest member that distrusts a byzantine member leaves; if a fraction a of honest members do so, the byzantine fraction rises from β to $\beta/(\beta + (1-\beta)(1-a))$. (3) Under seniority, the adversary can remove exactly the honest members that joined after some byzantine member. (4) Under scoped quorum, every honest member the adversary can remove, directly or through cascades, lies in $\partial \cup N(\partial)$.*

Proof. (1)–(3) are immediate. (4): removal requires at least one counted distruster; byzantine distrusters of honest x count only if they share a positive neighbour with x, which confines x to $\partial \cup N(\partial)$ as above; removals elsewhere lower $\deg_A(x)$ but do not change who lies within distance two of x. □

B.4 Signed conductance

For $S \subset A$ with $\mathrm{vol}_A(S), \mathrm{vol}_A(A \setminus S) > 0$, let

$$\Phi^{\pm}_A(S) = \frac{e^+(S, A \setminus S) - e^-(S, A \setminus S)}{\min\big(\mathrm{vol}_A(S), \mathrm{vol}_A(A \setminus S)\big)}, \qquad \Phi^{\pm}(A) = \min_S \Phi^{\pm}_A(S),$$

and let $\Phi^+(A)$ be the same with e^- omitted.

Assumption 3 (Awareness)**.** For a community A and $\kappa > 0$: $e^-(A \cap H \to A \cap B) \ge \kappa\, e^+(A \cap H, A \cap B)$.

Proof of Theorem 2. (1) For every S, $\Phi^{\pm}_A(S) \le \Phi^+_A(S)$, so $\Phi^{\pm}(A) \le \Phi^+(A)$ and any community admitted under a lower bound on $\Phi^{\pm}$ satisfies the same lower bound on Φ^+. A measure that may exceed Φ^+ admits communities the original theorem does not cover; Example 23 shows this can break the guarantee.

(2) Precisely: suppose $\alpha d \le \deg_A(v) \le d$ for $v \in A$, $e^+(A \cap H, A \cap B) \le \gamma \mathrm{vol}_A(A \cap H)$, awareness with $\kappa < 1$, and $\mathrm{vol}_A(A \cap B) \le \mathrm{vol}_A(A \cap H)$. If $\Phi^{\pm}(A) > (1-\kappa)\frac{\gamma}{\alpha} \cdot \frac{1-\beta}{\beta}$, then $\beta(A) < \beta$. Indeed, if $A \cap B = \emptyset$ there is nothing to prove. Otherwise take $S = A \cap B$; the minimum in the denominator is $\mathrm{vol}_A(S)$, and

$$\Phi^{\pm}(A) \le \frac{(1-\kappa)\, e^+(A \cap H, A \cap B)}{\mathrm{vol}_A(A \cap B)} \le \frac{(1-\kappa)\gamma\, d|A \cap H|}{\alpha d |A \cap B|} = (1-\kappa)\frac{\gamma}{\alpha} \cdot \frac{1-\beta(A)}{\beta(A)}.$$

If $\beta(A) \ge \beta$ the right side is at most $(1-\kappa)\frac{\gamma}{\alpha}\frac{1-\beta}{\beta}$, a contradiction.

(3) If $A \cap H$ and $A \cap B$ are both non-empty, take $S = A \cap B$. The numerator is at most $(1-\kappa)e^+(A \cap H, A \cap B) \le 0$ when $\kappa \ge 1$, and equals $-e^-(S, A \setminus S) \le 0$ when there are no attack edges inside A. Hence $\Phi^{\pm}(A) \le 0$. □

We checked parts (2) and (3) by exhaustive computation of net conductance on random small signed graphs in exact arithmetic.

B.5 Certified scarcity

Let $e^-(A)$ count distrust edges between members and $\rho^-(A) = e^-(A)/\mathrm{vol}_A(A)$.

Lemma 26 (Distrust-certified scarcity)**.** *Suppose $\alpha d \le \deg_A(v) \le d$ for $v \in A$, awareness holds for A, $|A \cap B| \le |A \cap H|$, and $\Phi^+(A) > \frac{2\rho^-(A)}{\kappa\alpha^2} \cdot \frac{1-\beta}{\beta}$. Then $\beta(A) < \beta$.*

Proof. If there is no attack edge inside A, then $\Phi^+(A) > 0$ makes $\Gamma^+|_A$ connected, and since $A \cap H \neq \emptyset$ there are no byzantine members. Otherwise $e^+(A \cap H, A \cap B) \le \kappa^{-1} e^-(A)$, and $\mathrm{vol}_A(A \cap H) \ge \alpha d|A \cap H| \ge \alpha d|A|/2 \ge \frac{\alpha}{2}\mathrm{vol}_A(A)$, so scarcity holds with $\gamma = 2\rho^-(A)/(\kappa\alpha)$, and the conductance condition is that of the single-community argument of [1] for this γ. □

C Layer 3: definitions and proofs

Disputes. A dispute opened by $\mathsf{Distrust}(u \to v)$ starts from the positive edge set E_0 after the declaration. Its state is (E_t, σ_t) with $E_t \subseteq E_0$ and $\sigma_t : V \to \{\mathsf{u}, \mathsf{v}, \circ\}$, initially u at u, v at v and $\circ$ elsewhere; the sides are $S_\mathsf{u}, S_\mathsf{v}$. With radius r, the *participants* are $P_r = \{w : d_{E_0}(w, \{u, v\}) \le r\}$ ($P_\infty = V$); only participants act. An unaligned participant is *exposed* to a side if it has an E_t-neighbour on it, and *conflicted* if exposed to both.

- $\mathsf{Align}(w, s)$, for w exposed to s: sets $\sigma(w) = s$ and removes the positive edges from w to the opposite side.
- $\mathsf{Abstain}(w)$, for conflicted w: removes the positive edges from w to both sides; w stays unaligned.

Under the conflict commitment, a run settles when no participant is conflicted and no participant wishes to act. With propagation, each edge $\{w, x\}$ removed by $\mathsf{Align}(w, \cdot)$ comes with $\mathsf{Distrust}(w \to x)$; abstention declares nothing.

The local commitment processes a declaration on $\{a, b\}$ as follows: each participant w that is a common positive neighbour of a and b removes one of $\{w, a\}, \{w, b\}$ (or both, if abstaining). With propagation, the removed edge $\{w, p\}$ becomes $\mathsf{Distrust}(w \to p)$ and opens a dispute on $\{w, p\}$; abstention opens none.

Lemma 27 (Invariants)**.** *In any run of the conflict commitment, with or without radius and abstention: (I1) no edge of E_t joins the two sides; (I2) every removed edge has one endpoint on a side, and its other endpoint is either on the opposite side or the abstaining identity; in particular edges between two unaligned identities are never removed; (I3) each side induces a connected subgraph of (V, E_t). There are at most $|V| - 2$ alignments and at most $|E_0|/2$ abstentions.*

Proof. Initially $\{u, v\} \notin E_0$. An alignment by w removes exactly its edges to the opposite side, which is unchanged; an abstention removes only edges from an unaligned w to side members. This gives (I1) and (I2). For (I3), an aligning w is exposed to its new side through an edge that is kept, and edges within a side are never removed. Each alignment fixes one identity; each abstention removes at least two edges. □

Proof of Theorem 5 and Theorem 8. Let K_r be the component of h in the subgraph of E_0 induced on $H \cap P_r$ (K_∞ is the component in $(H, E_0[H])$). (1) An honest participant acts only when obligated or willing. If conflicted it must act and, by loyalty, aligns with h's side, to which it is exposed; if exposed only to b's side it is neither obligated nor willing. (2) If w is honest it joins h's side, and the opposite side is byzantine by (1); if w is byzantine, every removed edge has endpoint w. (3)

By (2), edges between honest identities survive. Along a path $h = x_0, \ldots, x_k$ in K_r, if x_{i-1} is on h's side and x_i were unaligned, x_i would be a participant exposed to h's side and willing to align, contradicting settlement; so x_i is aligned, and by (1) on h's side. (4) follows from (3) and (I1). □

Proof of Corollary 6. Honest identities align only with h's side, so the identities they drop are on b's side, which is byzantine by Theorem 5(1). □

Proof of Proposition 7. For the conflict commitment, only participants act, and by (I2) every removed edge joins an acting participant to a side member, who is a participant. The count bounds the two balls of radius r. For the local cascade, by induction both parties of every dispute are participants: the original parties are, and a new dispute $\{w, p\}$ joins an acting participant w to a party p of a processed dispute. After the dispute on $\{a, b\}$ is processed, no participant is a common positive neighbour of a and b, and later steps only remove positive edges. □

Proof of Proposition 9. The bounds are those of Lemma 27. If every conflicted identity abstains and nobody aligns voluntarily, the sides remain $\{u\}$ and $\{v\}$; the conflicted identities are exactly the common neighbours of u and v, and each removes both its edges to them, after which nobody is conflicted. In the local cascade, an abstention removes edges without declaring distrust, so it opens no dispute. □

Proof of Proposition 10. An honest identity never aligns with b's side, so that side is byzantine. An honest identity that aligns joins h's side and removes edges to b's side, which have a byzantine endpoint; one that abstains is itself an endpoint of every edge it removes; a byzantine actor is an endpoint of every edge it removes. Under full loyalty a conflicted honest identity is exposed to h's side and joins it, so it never abstains. □

Proof of Theorem 3. Each dispute beyond the initial declarations D is opened by turning a positive edge negative, which happens at most once per edge, so there are at most $|D| + |E^+|$ disputes. After the dispute on $\{a, b\}$, a and b have no common positive neighbour, and later steps only remove positive edges; at the end every negative edge has been disputed, so no triangle has exactly one negative edge. If $(V, E^+ \cup D)$ is chordal, the underlying graph is fixed since edges only change sign; by induction on length no cycle has exactly one negative edge, because a chord splits a longer cycle into two shorter ones, and whichever sign the chord has, one of them has exactly one negative edge. Davis's theorem [8] gives clusterability with the positive components as clusters. □

Proof of Corollary 4. By induction, every dispute in the cascade is between an honest and a byzantine identity or between two byzantine identities. This holds initially. A new dispute $\{w, p\}$ is opened by w dropping p. If w is honest and the dispute it acted in was mixed, then by loyalty p is the byzantine party; if that dispute was between two byzantine identities, p is byzantine. If w is byzantine, the new dispute has a byzantine party. Hence an honest identity only ever drops, and declares distrust toward, byzantine identities. □

The statements in this appendix were also checked on thousands of random small instances, with no violation.

References

[1] O. Poupko, G. Shahaf, E. Shapiro, N. Talmon. Building a Sybil-resilient digital community utilizing trust-graph connectivity. *IEEE/ACM Trans. Networking* 29(5), 2021. arXiv:1901.00752.

[2] G. Shahaf, E. Shapiro, N. Talmon. Genuine personal identifiers and mutual sureties for Sybil-resilient community formation. arXiv:1904.09630, 2019.

[3] L. Cardelli, L. Orgad, G. Shahaf, E. Shapiro, N. Talmon. Digital social contracts: a foundation for an egalitarian and just digital society. arXiv:2005.06261, 2020.

[4] J. Golike, A. Lewis-Pye, E. Shapiro. Formalising and implementing grassroots social contracts: from legal text to grassroots platforms. arXiv:2609.06231, 2026.

[5] A. Lewis-Pye, E. Shapiro. Volition-guarded multiagent atomic transactions: describing people and their machines. arXiv:2604.25596, 2026.

[6] E. Shapiro. Grassroots platforms with atomic transactions: social networks, cryptocurrencies, and democratic federations. arXiv:2502.11299, 2025.

[7] F. Harary. On the notion of balance of a signed graph. *Michigan Math. J.* 2(2), 1953.

[8] J. A. Davis. Clustering and structural balance in graphs. *Human Relations* 20(2), 1967.

[9] T. Antal, P. L. Krapivsky, S. Redner. Dynamics of social balance on networks. *Phys. Rev. E* 72, 036121, 2005.

[10] J. Leskovec, D. Huttenlocher, J. Kleinberg. Signed networks in social media. *Proc. CHI*, 2010.